Spatio-Temporal Air Pollution Forecasting Using Graph Neural Networks and LSTM

Big Data Analytics (AI358) · B.Tech. Artificial Intelligence, 6th Semester · 2025–26

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Motivation & Problem Statement

PM2.5 is linked to ~7 million premature deaths annually (WHO), penetrating deep into the pulmonary system and bloodstream.

Why is This Hard?

Air quality forecasting is inherently **spatio-temporal**: pollution at any station is influenced by wind transport, topographic channeling, and proximity to emission sources — not just its own history.

Why GNN + LSTM?

Monitoring networks form natural graphs. GNNs aggregate spatial information across neighboring stations; LSTMs capture sequential temporal dependencies — together forming a principled ST-GNN architecture.

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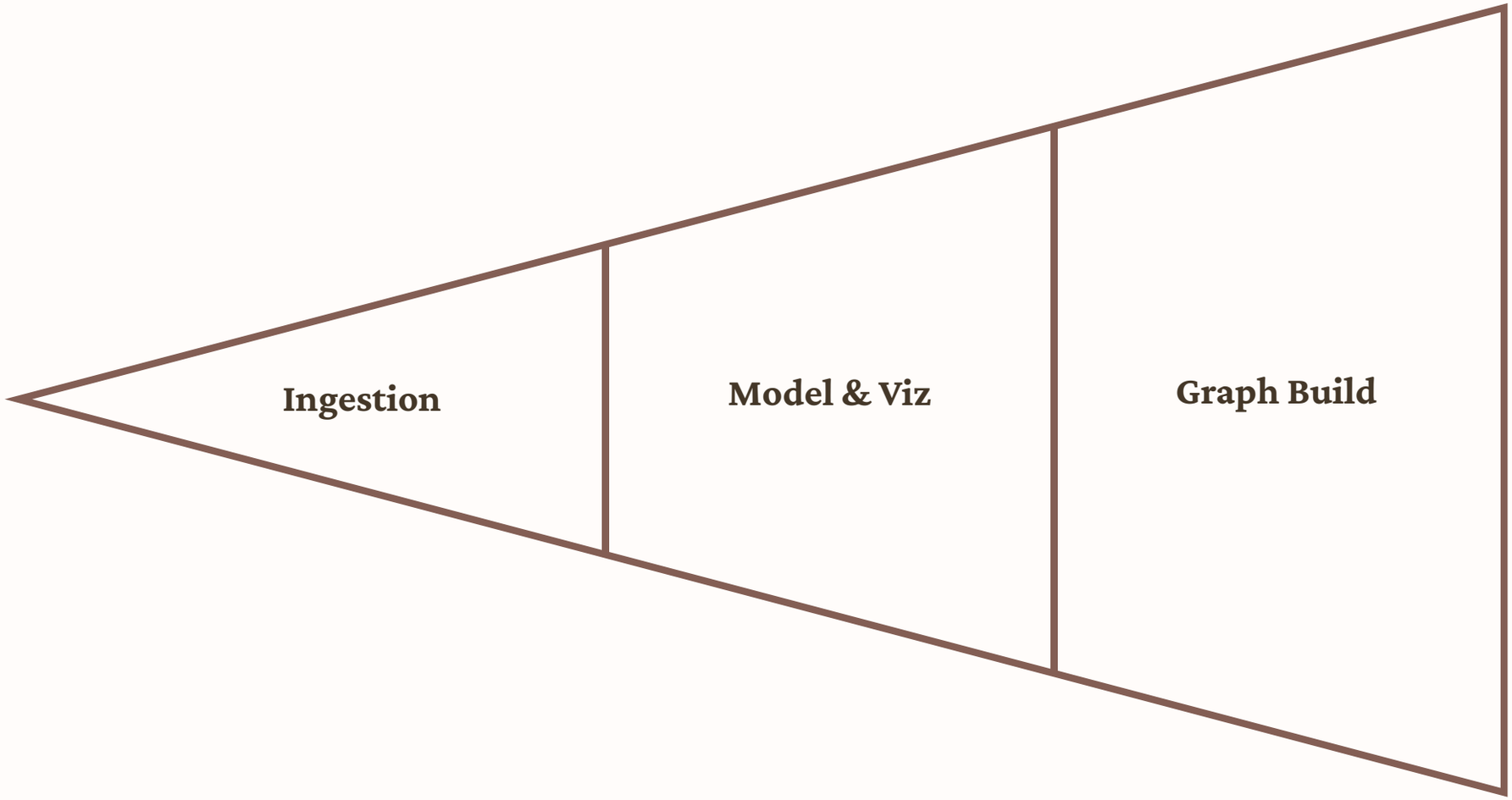
CHAPTER 1

Project Objectives & Pipeline at a Glance



Each stage persists intermediate artefacts — Parquet files, HBase tables, CSV exports — ensuring full reproducibility and enabling selective re-execution without reprocessing upstream data.

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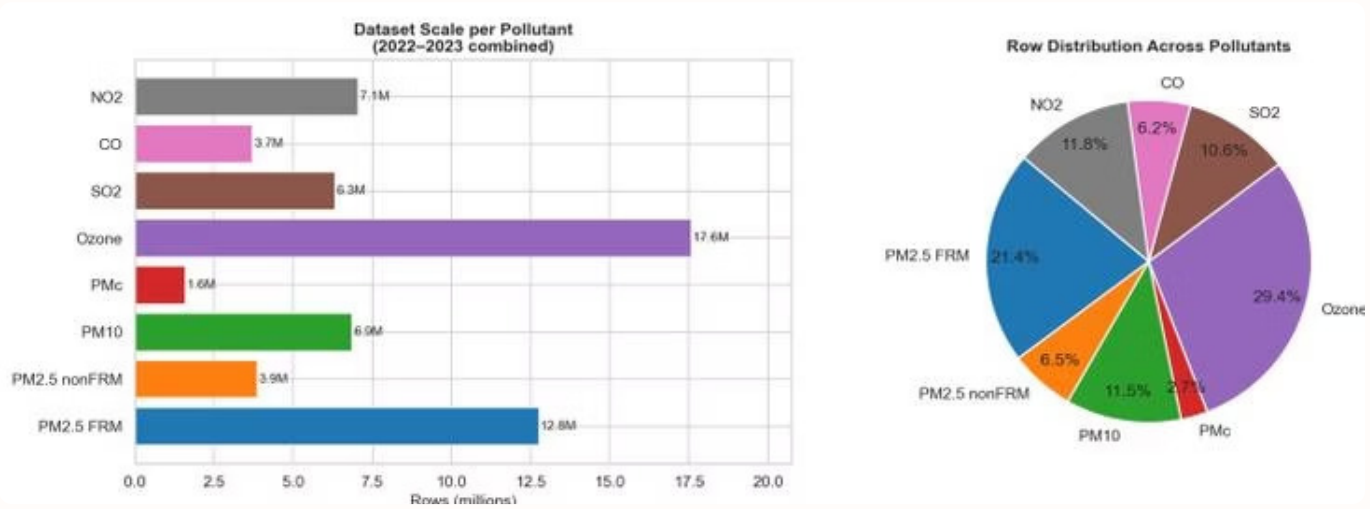
CHAPTER 2

Dataset: EPA Air Quality System (AQS)



Dataset Summary

Source	US EPA AQS
Period	2022–2023 (24 months, hourly)
PM2.5 Rows	12,489,096
Stations (post-filter)	805
PM2.5 Mean	8.33 μg/m³ (Std: 8.77)
PM2.5 Max	497.6 μg/m³ (wildfire events)
EPA 24h Standard	35.0 μg/m³

Ozone dominates at 17.4M rows (29.4% of all data); PM2.5 FRM constitutes 21.4% — the largest single regulated pollutant family by monitoring volume.

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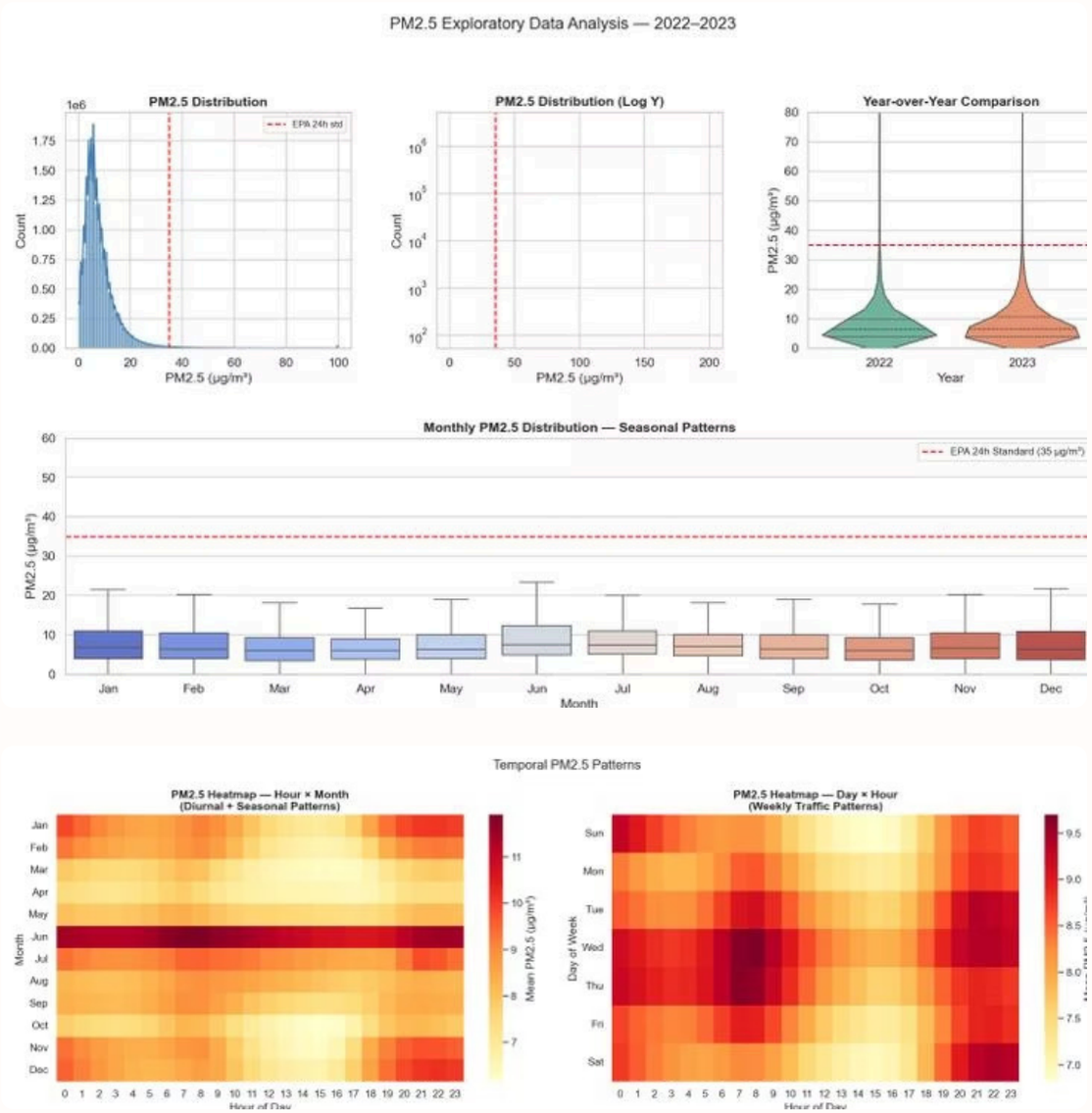
PM2.5 Distribution & Temporal Patterns



Distribution Characteristics

Concentrations follow a **right-skewed distribution**; extreme values reflect wildfire events and industrial incidents. The heavy tail motivates log-scale analysis and robust loss functions (Huber).

Temporal Cycles Exploited by LSTM

- **Diurnal:** Morning and evening traffic peaks, midday boundary-layer minimum
- **Seasonal:** Elevated winter concentrations (heating, low mixing height)
- **Weekly:** Weekday vs. weekend emission differences visible in hour×day heatmap

> Strong autocorrelation at 1h (r=0.960), 24h (r=0.629), and 168h (weekly) directly motivates the 12-hour LSTM input window.

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CHAPTER 3

Pipeline: Spark Ingestion, HBase Storage & Graph Construction

Stage 1 — PySpark Ingestion

10 GB driver memory; 100 shuffle partitions; adaptive query execution. Window functions add lag-1→6, 24h rolling mean/std, and time-of-day features. 5× Parquet compression vs. raw CSV.

Stage 2 — HBase Storage

Row key `station_id#YYYYMMDD#HH` enables sub-millisecond range scans. Three column families: `cf_raw`, `cf_features`, `cf_meta` — supporting both batch retrieval and online inference.

Stage 3 — Spatial Graph

BallTree-Haversine index; 150 km radius threshold; inverse-distance edge weights normalised per node. Result: **805 nodes, 10,928 directed edges**, mean degree 13.6 neighbours per station.

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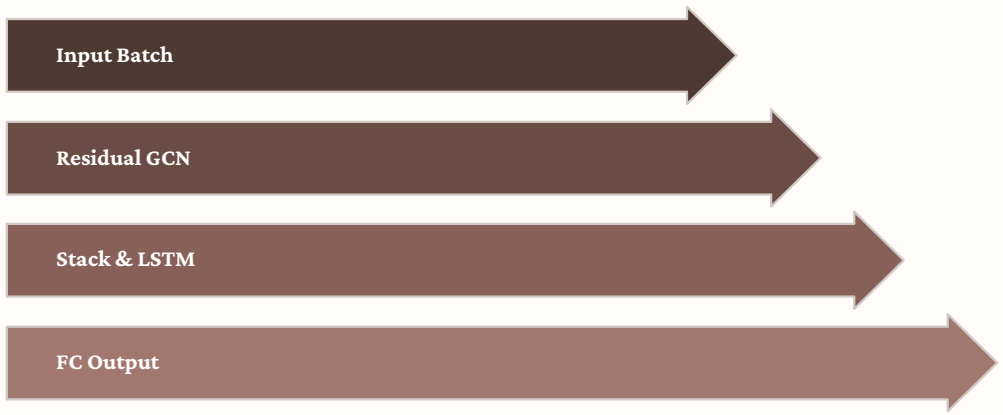
CHAPTER 4

ST-GNN-LSTM Model Architecture



Key Design Choices

Residual GCN

Skip connections prevent gradient vanishing; BatchNorm stabilises training across 805 heterogeneous stations. Hidden dim: 64.

2-Layer LSTM

Processes the 12-step spatially-enriched sequence per node; hidden dim 128 captures diurnal and multi-hour autocorrelation.

Huber Loss (δ=0.5)

Balances MSE sensitivity to spikes with MAE robustness — appropriate for the heavy-tailed PM2.5 distribution.

26,642,149 parameters · Trained on Apple M4 Pro (MPS) · 67.2 min · 9 epochs (early stopping)

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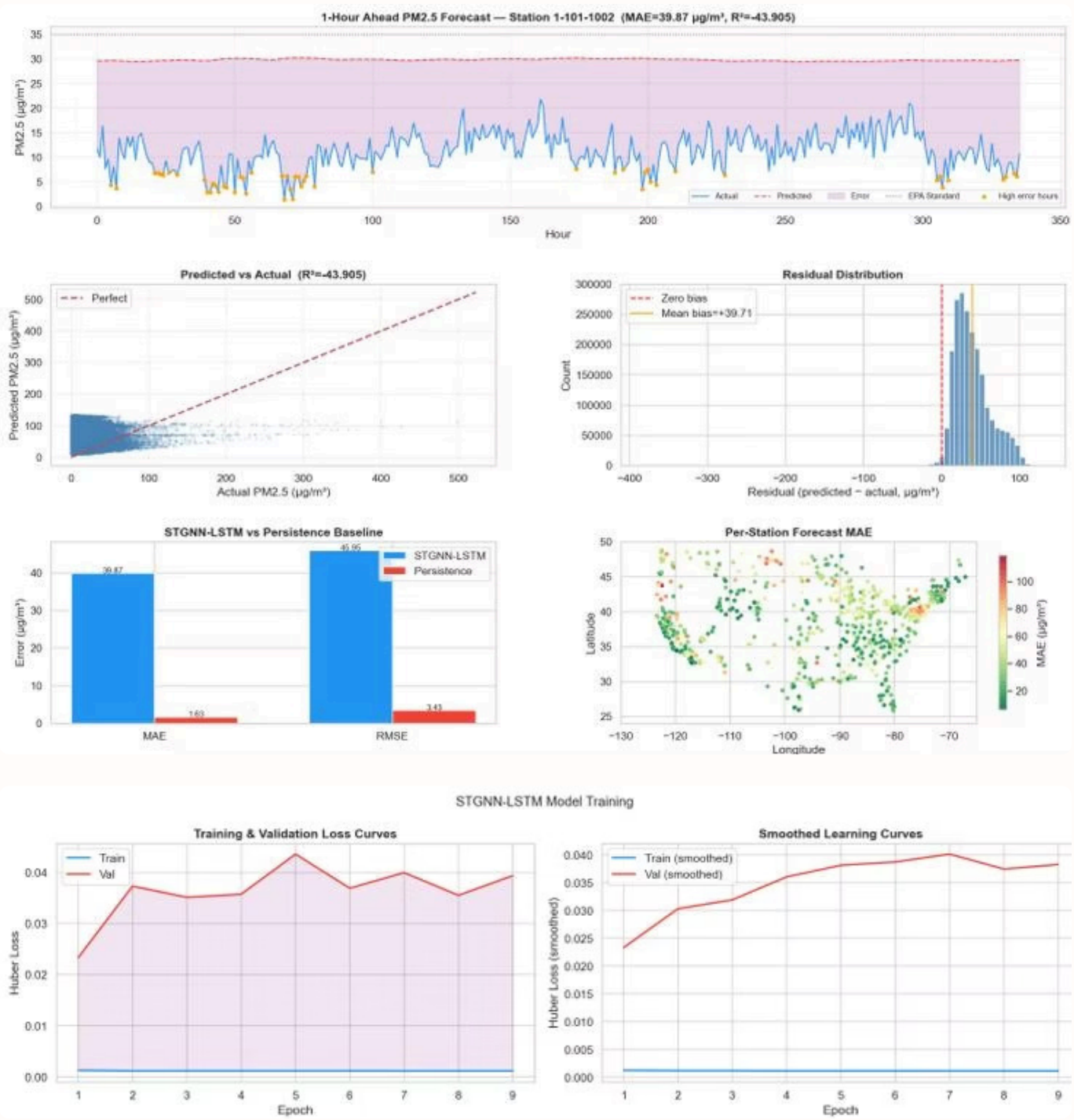
CHAPTER 5

Experimental Results



Test Set Metrics (15% held-out, chronological)

Metric	STGNN-LSTM
MAE	39.874 μg/m³
RMSE	45.953 μg/m³
Mean Bias	+39.714 μg/m³
Best Val Loss (Huber)	0.02336
Train Loss (Huber)	0.00116

⚠ The large train–validation gap indicates **distributional shift** at inference time — MinMaxScaler fitted on normalised training data does not generalise to the test distribution's scale. This is the primary limitation driving future work.

Validation loss oscillated between 0.025–0.038 across 9 epochs before early stopping triggered, indicating rapid macro-scale learning followed by difficulty generalising fine-grained temporal patterns.

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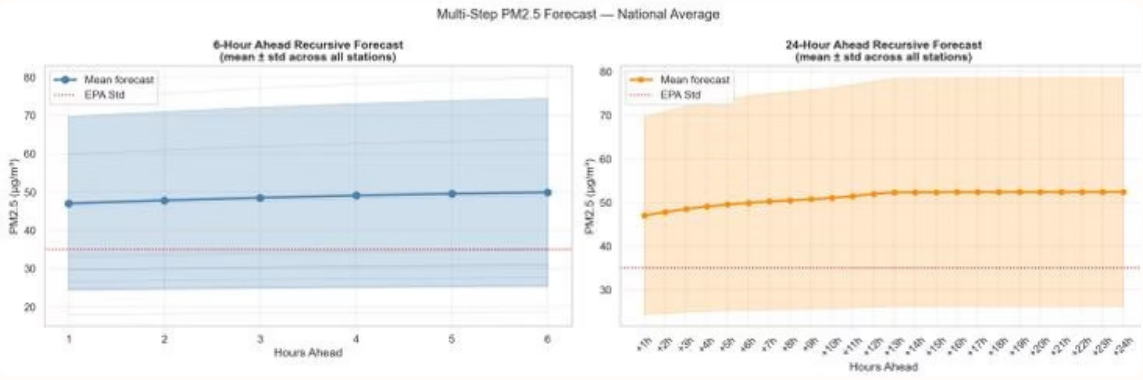
Multi-Step Forecasting & R Visualisation Suite



Recursive Multi-Step Forecasting

Predicted values are fed back as inputs for horizons beyond 1 hour. Forecast uncertainty propagates and widens with each step — a universal property of autoregressive models — visible in the shaded divergence zones at 6h and 24h horizons.

R Visualisation Output (8 Panels)

- **R1–R4:** Forecast vs. actual, training curves, 4-panel residual diagnostics, 24h multi-step forecast
- **R5–R6:** Diurnal ridge plots (ggridges) by month; state-level PM2.5 bar chart with exceedance annotations
- **R7–R8:** Interactive Leaflet map (stations coloured by per-station MAE); Plotly 3D surface of PM2.5 over geography and time

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Conclusions & Future Roadmap

✅ Key Contributions

- 12.4M-row PySpark pipeline with lag & rolling features
- HBase compound row-key for fast temporal range scans
- 805-node, 10,928-edge BallTree-Haversine spatial graph
- 26.6M-param ST-GNN-LSTM trained on Apple MPS in 67 min
- 8-panel R visualisation suite (static + interactive)

🔬 Limitations & Future Work

- **Scaler refitting** exclusively on training split to eliminate distributional shift
- **Graph Attention Networks (GAT)** for adaptive, timestep-dependent edge weighting
- **Multi-pollutant features**: O_3, NO_2, CO, and meteorological variables as node features
- **Teacher forcing / scheduled sampling** for improved multi-step accuracy
- **India CPCB dataset**: Apply pipeline to ~800 Indian stations (Ahmedabad AQI context)

The codebase, trained model checkpoint, and all exported figures are available on GitHub. The pipeline is fully extensible to additional years, pollutants, and geographic regions.

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